

DAVID SOLANILLAS

PhD Candidate | Robotics Simulation Researcher | Multibody Dynamics & Contact Mechanics

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Final-year PhD researcher at McGill University working on multibody systems with unilateral contact and friction, focused on robotics. Four first-author journal papers (RA-L, Mechanism and Machine Theory, ASME Journal of Applied Mechanics). I have implemented and validated position-based dynamics algorithms in C++ inside CM Labs' Vortex dynamics engine. I am currently focused on how contact-model fidelity limits what robot policies can learn in simulation.

EDUCATION

PhD in Mechanical Engineering | 2021 – Present

McGill University, Montreal, Canada

MSc in Mechanical Engineering | 2018 – 2020

University of the Basque Country, Bilbao, Spain

Ranked 1st out of 14 graduates. Multiple Honours.

BSc in Mechanical Engineering | 2014 – 2018

University of Zaragoza, Zaragoza, Spain

Ranked 7th out of 124 graduates (top 6%). Completed in 4 years against a 5.7-year cohort average. Multiple Honours

RESEARCH & ENGINEERING EXPERIENCE

PhD Researcher – Applied Dynamics Lab, McGill University | 2021-Present

Supervisor: Prof. József Kövecses

- Developed dynamic formulations and time-stepping schemes for multibody systems with unilateral contact and friction, applied to robotic manipulators.
- Derived a point-mass formulation for rigid-body systems in which each link is modelled by four point masses, giving a constant mass matrix, no nonlinear inertial terms, and constraint Jacobians that are linear in the generalized coordinates.
- This is particularly advantageous for differentiable physics, where the expensive terms either vanish or can be precomputed ($\partial M / \partial q = 0$, $\partial c / \partial q = 0$, constant Hessians for the constraint Jacobians).
- Developed a new method for computing the Coriolis/centrifugal matrix for open- and closed-loop manipulators (IEEE RA-L, 2026).

Contact Modelling and Robot Learning | 2021-Present

- Designed benchmark tests comparing our own LCP-based bristle-friction models against MuJoCo on robotic manipulation tasks, including grasping and non-prehensile tasks.
- Traced the differences to the formulation: relaxed complementarity in convex models does not lead to physically accurate representations of stick-slip transitions or true sticking. A published sim-to-real benchmark required adding a virtual wall to make the same task trainable, while our LCP-based bristle-friction model reproduced it using friction alone.
- Co-authored talk presented at RoboSim 2026 (London, UK); to be presented at the IROS 2026 Workshop on Contact & Learning (Pittsburgh, USA).

Research Collaboration – CM Labs Simulations, Inc. (Vortex Studio), Montreal, Canada | 2025 - Present

- Implemented and validated position-based dynamics algorithms in C++ within the Vortex commercial simulation engine, integrating them into an existing production codebase.

Researcher – University of the Basque Country, Bilbao, Spain | 2020-2021

- Modelled and validated compliant closed-loop manipulators, analyzing their kinematics and dynamics.
- The research led to a peer-reviewed journal publication.

Researcher – Tekniker, Eibar, Spain | 2019-2020

- Finite element analysis of machine-tool electrospindles, including experimental modal analysis and thermal characterization.

TECHNICAL SKILLS

MATLAB – Primary research language, used to implement the dynamic formulations and time-stepping schemes behind all my publications.

C++ – Used to write the position-based dynamics methods developed for our research partner, CM Labs Simulations.

Python – MuJoCo benchmarking and analysis for the contact-fidelity studies.

CUDA – NVIDIA DLI certification, Fundamentals of Accelerated Computing with CUDA C/C++ (2022).

Simulation and numerical methods – non-smooth dynamics; unilateral contact and friction modelling; LCP/MLCP formulations and pivoting methods; bristle friction models and convex relaxations of the contact problem; time-stepping and integration schemes; position-based dynamics; differentiable physics.

Tools – CM Labs Vortex, MuJoCo.

Languages – Spanish (native), English (professional), French (intermediate).

ACADEMIC SERVICE

- Reviewer, *Mechanism and Machine Theory* (Elsevier).

AWARDS AND RECOGNITION

- International McGill Engineering Doctoral Award (MEDA International), 2021.
- McGill Graduate Excellence Award in Engineering, 2021-2025.
- Internal Graduate Studies Fellowship, University of the Basque Country, 2018-2020.
- Undergraduate and Graduate Studies Fellowship, Ministry of Education, Culture and Sports of Spain, 2014-2020.

LIST OF PUBLICATIONS

- D. M. Solanillas Francés and J. Kövecses, "A New Approach for the Computation of the Coriolis/Centrifugal Matrix", *IEEE Robotics and Automation Letters*, Vol. 11(10), pp. 11442-11449, 2026. [\[DOI\]](#)
Eligible for presentation at ICRA/IROS 2027.
- D. M. Solanillas Francés and J. Kövecses, "Dynamics Representation of Mechanical Systems for Time-Stepping Problems", *Mechanism and Machine Theory*, Vol. 218, pp. 106292, 2025. [\[DOI\]](#)
- D. M. Solanillas Francés and J. Kövecses, "A New Representation for the Dynamics of Rigid Bodies", *ASME Journal of Applied Mechanics*, Vol. 92 (7), pp. 071004, 2025. [\[DOI\]](#)
- D. M. Solanillas Francés and J. Kövecses, "A New Formulation for the Dynamics of Rigid Bodies with Unilateral Interactions", *Mechanism and Machine Theory*, Vol. 203, pp. 105809, 2024. [\[DOI\]](#)
- O. Altuzarra, D. M. Solanillas, E. Amezua and V. Petuya, "Path Analysis for Hybrid Rigid-Flexible Mechanisms", *Mathematics*, Vol. 9(16), pp. 1869, 2021. [\[DOI\]](#)

LIST OF CONFERENCES

- D. M. Solanillas Francés and J. Kövecses, "A Novel Dynamics Formulation for Articulated Mechanical Systems with Unilateral Contacts", *12th Asian Conference on Multibody Dynamics (ACMD)*, October 25-29, 2026. Okinawa, Japan. (Upcoming).
- D. M. Solanillas Francés and J. Kövecses, "Contact Modelling Still Matters for Learning-Based Manipulation", *Workshop on Contact & Learning, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, September 27-October 1, 2026. Pittsburgh, PA, USA. [\[Online\]](#). (Poster, upcoming).

- D. Solanillas, C. Sirois, S. Arbatani and J. Kövecses, "Accurate Contact and Friction Modelling: Importance in Robotic Simulation", **RoboSim Summit**, September 15-17, 2026. London, United Kingdom.
- D. M. Solanillas Francés and J. Kövecses, "Particle-Based Simulation of Rigid Bodies for Real-Time Applications", *17th World Congress on Computational Mechanics & 10th ECCOMAS Congress (WCCM-ECCOMAS)*, July 19-24, 2026. Munich, Germany. [\[Online\]](#)
- D. M. Solanillas Francés and J. Kövecses, "A Two-Stage Dynamics-Kinematics Representation for Unilateral Contact in Multibody Systems", *8th International Conference on Multibody System Dynamics (IMSD)*, June 16-19, 2026. Seville, Spain.
- D. M. Solanillas Francés and J. Kövecses, "A Novel Dynamics Representation for Unilateral Contact Problems", *Contact Mechanics International Symposium (CMIS)*, April 21-24, 2026. Lugano, Switzerland. (Poster).
- D. M. Solanillas Francés and J. Kövecses, "Fragmentation and Demolition Simulation Using Particle Systems", *9th ECCOMAS International Conference on Particle-Based Methods (ECCOMAS Particles)*, October 20-22, 2025. Barcelona, Spain. [\[Online\]](#)
- D. M. Solanillas Francés and J. Kövecses, "A Novel Time-Stepping Approach for Mechanical Systems", *12th ECCOMAS Thematic Conference on Multibody Dynamics (ECCOMAS Multibody)*, July 13-18, 2025. Innsbruck, Austria. [\[Online\]](#)
- D. M. Solanillas Francés and J. Kövecses, "Eulerian-Based Rigid Body Dynamics: A New Perspective", *8th ECCOMAS International Conference on Computational Contact Mechanics (ECCOMAS ICCCM)*, July 2-4, 2025. Munich, Germany.
- D. M. Solanillas Francés and J. Kövecses, "Point Mass Based Formulation for Unilateral Contact Problems", *ASME International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE)*, August 25-28, 2024, Washington, DC, USA.
- D. M. Solanillas and J. Kövecses, "A Novel Approach for Mechanical Systems with Unilateral Interactions", *11th European Nonlinear Dynamics Conference (ENOC)*, July 22-26, 2024, Delft, The Netherlands. [\[Online\]](#)
- D. M. Solanillas Francés and J. Kövecses, "Alternative Formulation for Modelling Rigid Bodies in Unilateral Interaction Problems", *Mechanism and Machine Theory Symposium – Celebrating 60 years since the journal's foundation*, June 26-28, 2024, Guimarães, Portugal. [\[Online\]](#)
- D. M. Solanillas and J. Kövecses, "A Novel Formulation for Multibody Systems with Unilateral Interactions Using Equipomental Systems of Point Masses", *7th International Conference on Multibody System Dynamics (IMSD)*, June 9-13, 2024. Madison, WI, USA. [\[Online\]](#)